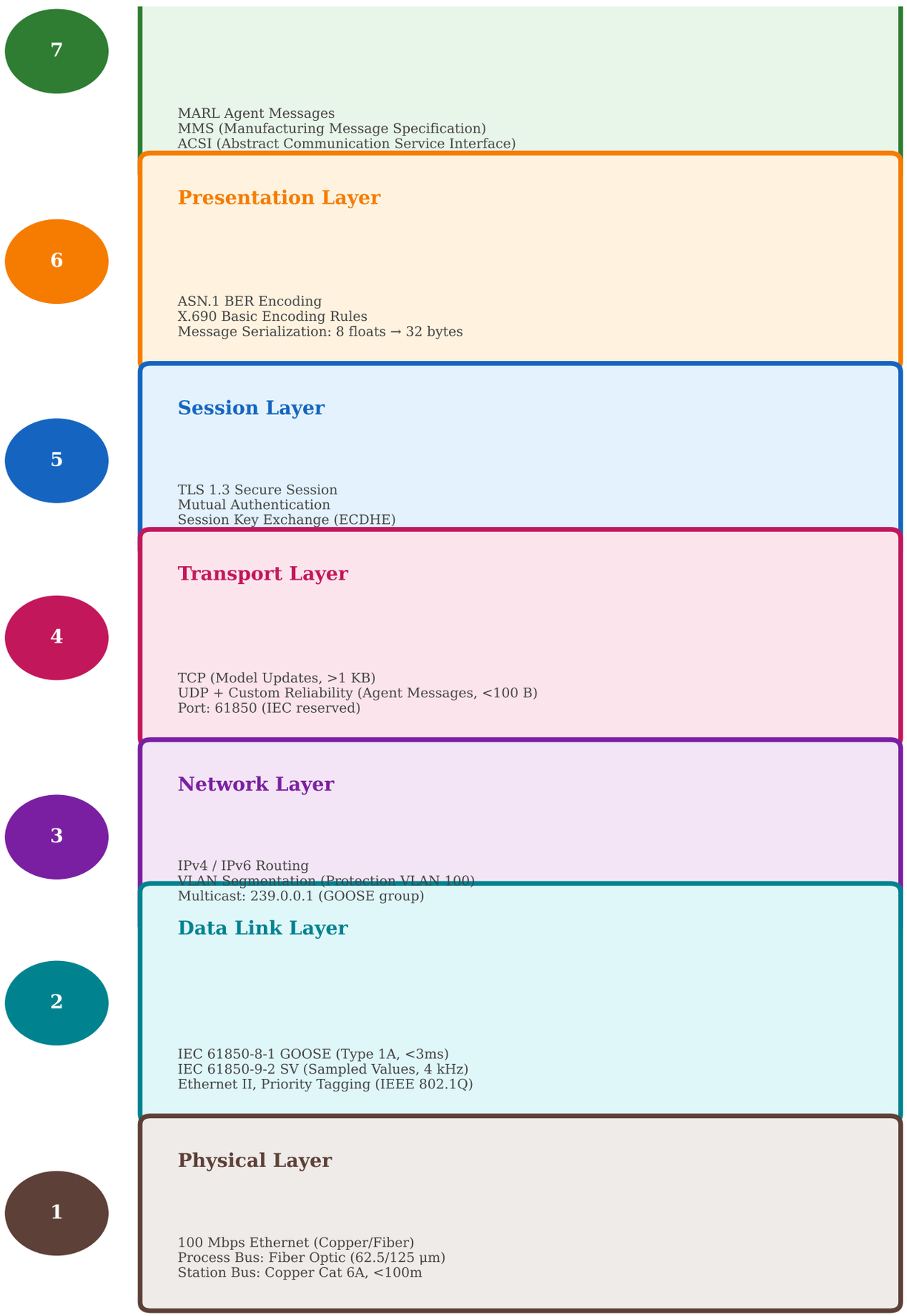
# Figure 51: Communication Protocol Stack — OSI 7-Layer Mapping for GNN Message Passing onto IEC 61850




## GNN Message → IEC 61850 Mapping

### GNN Message Structure (32 bytes):

| | | |
|---|---|---|
| **Bytes 0–3** | [float32] | Feature 1: Local V_mag (normalized) |
| **Bytes 4–7** | [float32] | Feature 2: Local I_mag (normalized) |
| **Bytes 8–11** | [float32] | Feature 3: P_dir (−1, 0, 1 encoded) |
| **Bytes 12–15** | [float32] | Feature 4: I_fault (log-compressed) |
| **Bytes 16–19** | [float32] | Feature 5: PV generation level (%) |
| **Bytes 20–23** | [float32] | Feature 6: Topology flag (0/1) |
| **Bytes 24–27** | [float32] | Feature 7: Neighbor TMS (avg) |
| **Bytes 28–31** | [float32] | Feature 8: Neighbor PS (avg) |

### GOOSE Message Mapping (Setting Updates):

- stVal (BOOLEAN): Setting update trigger
- f (FLOAT32): New TMS value (×1000 for precision)
- f (FLOAT32): New PS value (×100 for precision)
- q (Quality): Validity flag + timestamp
- T0–T3: Retransmission sequence (2, 4, 8, 16 ms)

### SV (Sampled Values) Mapping (Measurements):

- i (INT32): Instantaneous current (1 mA resolution)
- v (INT32): Instantaneous voltage (10 mV resolution)
- SmpCnt: Sample counter (0–3999 for 4 kHz)
- SmpSynch: Synchronization status (IEEE 1588 PTP)

### MMS Mapping (Configuration & Monitoring):

- GetDataValues: Read current relay settings (TMS, PS)
- SetDataValues: Write new settings (authenticated)
- Report (URCB): Unbuffered report on data change
- FileTransfer: Model update (ONNX binary, ~2 MB)

### End-to-End Performance Metrics:

- GNN message latency: < 3 ms (GOOSE T0 retransmission)
- SV measurement delay: < 250 µs (4 kHz sampling)
- Total inference-to-action: 45 ms (mean) | 82 ms (95th percentile)

### Bandwidth Budget:

- Per-agent GNN: 0.256 kbps (8 floats × 4 bytes × 10 Hz)
- GOOSE overhead: ~0.05 kbps (setting updates, ~1 per minute)
- SV reception: ~64 kbps (4 kHz × 8 bytes × 2 channels)
- **Total per-agent: ~64.3 kbps << AMI/RF mesh limit (1 Mbps)**